

P6UC04: Edit Equipment Information

Overview

****Actor:**** Company Owner, Manager, Staff

****Preconditions:****

- User is authenticated and has active session
- User has permission to edit equipment (Company Owner, Manager, or Staff role)
- Equipment item exists and belongs to user's tenant
- Equipment is not soft-deleted

****Postconditions:****

- Equipment information is updated
- updated_at timestamp is refreshed
- updated_by field is set to current user
- Audit log entry is created
- User receives success confirmation

****Trigger:**** User clicks "Edit" button from equipment detail page (UC6.3) or equipment list (UC6.2)

Main Flow

1. System retrieves equipment record by ID and validates it belongs to user's tenant
2. System retrieves active categories for category dropdown
3. System displays equipment edit form pre-populated with current data:
 - ****Basic Information:****
 - Equipment name (text input, required, max 200 characters, current value)
 - Equipment category (dropdown, required, populated from active categories, current selection highlighted)
 - Description (textarea, optional, max 2000 characters, current value)
 - ****Specifications:****
 - Manufacturer (text input, optional, max 100 characters, current value)
 - Model/Model number (text input, optional, max 100 characters, current value)
 - Serial number (text input, optional, max 100 characters, current value)
 - Year of manufacture (number input, optional, range: 1900 to current year, current value)
 - ****Financial Details:****
 - Purchase price (number input, optional, currency format, current value)
 - Purchase date (date picker, optional, cannot be in future, current value)
 - ****Status:****

- Status (dropdown, required, options: Available, Rented, Maintenance, Out of Service, current selection highlighted)
 - ****Custom Attributes:****
 - Existing custom attributes displayed as editable key-value pairs
 - "Add Attribute" button to add new attributes
 - "Remove" button next to each attribute
 - ****Read-only fields:****
 - Equipment ID (displayed but not editable)
 - Created by and created at (displayed but not editable)
4. User modifies desired fields
 5. User optionally manages custom attributes by:
 - Editing existing attribute key, value, or unit
 - Adding new attributes via "Add Attribute" button
 - Removing attributes via "Remove" button
 6. User submits the form
 7. System validates all inputs (see Alternative Flow A1 for validation failures)
 8. System verifies serial number uniqueness within tenant if changed (see Alternative Flow A2 for duplicate)
 9. System checks for optimistic locking conflicts (see Alternative Flow A3 for concurrent update)
 10. System updates equipment record in equipment_db with:
 - All modified data
 - updated_by = current user ID
 - updated_at = current timestamp (UTC)
 11. System stores updated custom attributes as JSON structure
 12. System creates audit log entry with:
 - action: 'EQUIPMENT_UPDATED'
 - user_id: current user ID
 - tenant_id: current tenant ID
 - entity_id: equipment ID
 - metadata: changes made (old values and new values)
 13. System displays success message in user's preferred language
 14. System redirects user to equipment detail view with updated information

Alternative Flows

A1: Validation Failure

****Trigger:**** Step 7 - User input fails validation rules

1. System identifies validation errors:
 - Equipment name is empty or exceeds 200 characters
 - Equipment category is not selected
 - Description exceeds 2000 characters

- Manufacturer exceeds 100 characters
 - Model exceeds 100 characters
 - Serial number exceeds 100 characters
 - Year of manufacture is in the future
 - Purchase date is in the future
 - Status is not selected from valid enum values
 - Custom attribute key/value exceeds length limits
2. System highlights fields with errors
 3. System displays validation error messages near respective fields in user's preferred language
 4. User corrects the errors
 5. Flow returns to Step 6

A2: Duplicate Serial Number

****Trigger:**** Step 8 - Modified serial number already exists for another equipment item within tenant

1. System detects serial number conflict
2. System displays error message: "This serial number is already assigned to another equipment item"
3. User modifies serial number to unique value or clears it
4. Flow returns to Step 6

A4: Selected Category Becomes Inactive

****Trigger:**** Step 8 - User changes category to one that became inactive after form load

1. System detects selected category is no longer active
2. System displays error message: "The selected category is no longer active. Please select another category."
3. System refreshes category dropdown with current active categories
4. User selects a different active category
5. Flow returns to Step 6

A5: Equipment Has Active Bookings (Future Constraint)

****Trigger:**** Step 8 - User attempts to change category for equipment with active rental bookings (placeholder for now - future phase validation)

1. System detects equipment has active or future rental bookings
2. System displays error message: "Cannot change category for equipment with active or future bookings"
3. Category field is disabled or reverted to original value
4. User can modify other fields but not category
5. Flow returns to Step 6

A6: User Cancels Edit

****Trigger:**** User clicks "Cancel" button at any step before submission

1. System prompts user: "Are you sure you want to discard unsaved changes?"
2. If user confirms cancellation:
 - System discards all changes
 - System redirects user to equipment detail view
3. If user declines cancellation:
 - System returns to edit form with all changes preserved

A7: Equipment Not Found or Unauthorized

****Trigger:**** Step 1 - Equipment ID does not exist, belongs to different tenant, or user lacks edit permission

1. System detects unauthorized access or missing equipment
2. System displays error message: "Equipment not found or you don't have permission to edit it"
3. System provides "Back to Equipment List" button
4. User clicks button
5. System redirects user to equipment list page (UC6.2)

A8: Equipment Is Soft-Deleted

****Trigger:**** Step 1 - Equipment record has deleted_at timestamp

1. System detects equipment is archived/soft-deleted
2. System displays error message: "Cannot edit archived equipment. Please restore it first." (placeholder for now - restore functionality)
3. System provides "Back to Equipment List" button
4. User clicks button
5. System redirects user to equipment list page (UC6.2)

A9: Session Expires During Edit

****Trigger:**** User's session expires while editing equipment

1. System detects expired session upon form submission
2. System redirects user to login page
3. After successful login, system redirects user back to equipment edit form (placeholder for now - depends on session recovery implementation)
4. User must re-enter changes (unsaved data is lost)

Business Rules

****BR6.4.1:**** Users can only edit equipment belonging to their tenant (strict tenant isolation)

****BR6.4.2:**** Only users with Company Owner, Manager, or Staff roles can edit equipment

****BR6.4.3:**** Equipment ID is system-generated and cannot be modified

****BR6.4.4:**** Creation metadata (created_by, created_at) cannot be modified

****BR6.4.5:**** Serial number must remain unique within tenant scope if changed

****BR6.4.6:**** Only active categories can be assigned during edit

****BR6.4.7:**** Year of manufacture cannot be greater than current year

****BR6.4.8:**** Purchase date cannot be in the future

****BR6.4.10:**** updated_at timestamp is automatically set to current time on save

****BR6.4.11:**** updated_by field is automatically set to current user ID

****BR6.4.12:**** Audit log must capture all changes with old and new values

****BR6.4.13:**** Custom attributes support flexible schema and can be added/removed freely

****BR6.4.14:**** Soft-deleted equipment cannot be edited until restored

****BR6.4.15:**** All timestamps are stored in UTC

****BR6.4.16:**** Form labels, validation messages, and success messages respect user's language preference (English or Persian)

****BR6.4.17:**** Equipment with active rental bookings cannot have category changed (future phase constraint - placeholder for now)